Lesson 6.6 Guided Notes

Session 1 · Simplifying to Show Equivalence

I can simplify an expression by combining like terms to test whether two expressions are equivalent. · 6.EE.A.3

Word / Spanish	Meaning — complete the blank	Example
like terms / términos semejantes	Terms with the same _____ part.	7c and 6c
combine / combinar	Add or subtract like terms into _____ term.	7c + 6c = 13c
equivalent expressions / expresiones equivalentes	Expressions that have the same _____ for every number.	7c + 9 + 6c and 13c + 9

Key idea

Like terms have the same _____ part; 4x and 4 are not like terms.

Combine like terms by adding their _____: 4x + 9x = 13x.

Two expressions are equivalent if they simplify to the _____ expression.

Worked example

Simplify 9b + 4 + 3b + 1.

The like terms with b are 9b and _____.

Combine them: 9b + 3b = _____.

Combine the constants: 4 + 1 = _____.

Answer: 9b + 4 + 3b + 1 = _____

You try

1. Simplify.
a. 6a + 2a = _____
b. x + 4x + 3 = _____

_____

2. Are 2n + 7 + 3n and 5n + 7 equivalent? Simplify to decide.

_____

Sum it up: I can combine 3x and 5x but not 3x and 5 because ___.

_____

Session 2 · Testing and Justifying Equivalence

I can test whether two expressions are equivalent by substituting values and by reasoning. · 6.EE.A.4

Word / Spanish	Meaning — complete the blank	Example
equivalent expressions / expresiones equivalentes	Expressions with the same value for _____ value of the variable.	2(y + 7) and 2y + 14
counterexample / contraejemplo	One test value that shows a claim is _____.	x = 1: 2x + 8 = 10, 4x = 4
Distributive Property / propiedad distributiva	Multiply the number outside by _____ term inside the parentheses.	4(n + 3) = 4n + 12
Commutative Property / propiedad conmutativa	You can change the _____ when you add or multiply.	a + b = b + a

Key idea

To test, substitute the _____ value into both expressions.

One value that gives different results proves they are _____ equivalent.

Matching values do not prove it; _____ with properties.

Worked example

Are 2x + 8 and 4x equivalent?

Try x = 4: 2 × 4 + 8 = 16 and 4 × 4 = _____. One match is not proof.

Try x = 1: 2 × 1 + 8 = _____ and 4 × 1 = 4.

10 ≠ 4, so x = 1 is a _____.

Answer: They are _____ equivalent.

You try

1. Evaluate when a = 3.
a. 2(a + 1) = _____
b. 2a + 1 = _____

_____

2. Are 5m and m + 4 equivalent? Test m = 1 and m = 2.

_____

Sum it up: One matching test value does not prove two expressions are equivalent because ___.

_____

Session 1 · Model, reason, and explain

Simplifying to Show Equivalence · 6.EE.A.3

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use a grouping or tile model to simplify 4x + 3 + 2x + 5. Explain why 4x and 2x combine but 4x and 3 do not.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: combines variable terms and constants into one coefficient.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Testing and Justifying Equivalence · 6.EE.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Are 3(x+2) and 3x+6 equivalent? Are 3x+6 and 6x equivalent? Give a property argument for one and a counterexample for the other.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: treats one matching substitution as proof of equivalence.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?